

HARIOM THACKER

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SUMMARY

Electronics and Communication Engineering undergraduate with hands-on experience across full-stack development (Node.js, Express, React, MySQL), data structures and algorithms, and applied machine learning/computer vision. Ranked 2nd globally at the NTIRE 2026 CVPR Workshop among 44 international teams. Seeking a Technology Internship to apply strong problem-solving and software engineering fundamentals.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT)

Surat, India

B.Tech, Electronics and Communication Engineering — **CGPA: 8.76/10.0**

2024 – 2028

- **Coursework:** Data Structures & Algorithms, OOP, Operating Systems, C/C++.

RESEARCH PROJECTS

HAFT — Aperture-Conditioned Bokeh Rendering

Jan 2026 – Present

NTIRE 2026 CVPR Workshop Challenge — Team Project — [GitHub]

- **Ranked 2nd overall / 1st in perceptual quality (LPIPS 0.0841)** among 44 international teams, architecting a Hybrid Aperture-conditioned Feature Transformer (HAFT) with a differentiable attention mechanism learned end-to-end from reconstruction loss.
- **Optimized inference to 0.42s/image** (RTX A5000) for +0.10 dB PSNR over the strongest baseline checkpoint, adding only +0.17M parameters.

HR Depth Estimation — Compute-Constrained Optimization

Jan 2025 – Feb 2026

NTIRE 2026 CVPR Workshop Challenge — Team Project — [GitHub]

- **Cut RMSE by 3.1% at zero extra training cost** via an optimized patch-tiling preprocessing system and log-space test-time augmentation, ranking 9th globally under the tightest compute budget in the challenge (48h, 2x T4 GPUs).

SELECTED PROJECTS

TransitOps — Full-Stack Fleet Operations Console

2026

Odoo Hackathon 2026 — Node.js, Express, MySQL, React — [GitHub]

- **Built a 13-rule dispatch validation engine and atomic state machine** for fleet dispatch/completion/cancellation, with live KPI analytics and maintenance locking on an Express + MySQL backend.
- **Designed for standalone demo or live backend** via a client-side rules engine that mirrors full server logic behind a single environment flag.

Pixify: Low-Latency AI Image Detector

Dec 2025 – Feb 2026

Genesis Deep Learning Competition (Top 3) — [GitHub]

- **Achieved 99.5% test accuracy** with a dual-input FFT-CNN processing spatial RGB and 2D DFT magnitude spectra to flag synthetic image artifacts; compiled via TorchScript for real-time CPU inference in a deployed Streamlit app.

From-Scratch Transformer & GAN Implementations

Nov 2025 – Dec 2025

Independent Study — [GitHub]

- **Built an English-to-Italian seq2seq Transformer from scratch** (multi-head attention, layer norm) and 4 GAN variants (Vanilla, WGAN, WGAN-GP, CGAN), implementing gradient-penalty and Wasserstein loss without high-level wrapper APIs.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL

CS Core: Data Structures, Algorithms, Operating Systems

Web & Backend: Node.js, Express, React, MySQL, REST APIs, Git

ML/DL: PyTorch, TorchScript, OpenCV, NumPy, Pandas, Scikit-learn

HONOURS & AWARDS

Rank 2 Overall + Rank 1 LPIPS — NTIRE 2026 CVPR Workshop (44 global teams)

2026

Rank 9 Overall — NTIRE 2026 CVPR Workshop (Most constrained compute budget)

2026

Top 3 Finish — Genesis AI Competition

2026

Top 25 National Finalist — GDG Autonomous Hacks

2026